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KFP-YOLO: A Lightweight Detection Model for Korla Fragrant Pear Disease and Pest Detection Toward Edge Deployment

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Abstract

Korla fragrant pear disease and pest detection faces challenges such as significant object scale variation, multi-organ target confusion, and limited computational resources for real-time inference on edge devices. This study presents KFP-YOLO, a lightweight object detector based on YOLO26n, and constructs the Korla Fragrant Pear Disease and Pest Dataset (KFP-PDD), which covers leaves, fruits, and flowers and contains 14,092 original images, 22,735 annotated instances, and 11 healthy, disease, and pest categories. In KFP-YOLO, ADown modules are introduced into the backbone to reduce downsampling redundancy, selected C3k2 blocks are replaced with C3-PD modules combining partial convolution and squeeze-and-excitation attention, and selected feature-fusion nodes are redesigned as CFA modules incorporating coordinate attention. Under identical training and evaluation settings, KFP-YOLO reduces the number of parameters from 2.51 M to 1.93 M and the computational complexity from 5.79 to 4.42 GFLOPs, corresponding to reductions of 23.1% and 23.7%, respectively. On the KFP-PDD test set, KFP-YOLO achieves an mAP@0.5 of 0.9429 and an mAP@0.5:0.95 of 0.7434, compared with 0.9458 and 0.7579 for the YOLO26n baseline. Its inference speed reaches 278.97 FPS on the Jetson AGX Orin platform. In addition, evaluation on an independent external test set containing 1052 images and 3377 annotated instances yields a Precision of 0.8814, a Recall of 0.9077, an mAP@0.5 of 0.9325, and an mAP@0.5:0.95 of 0.7322. These results indicate that KFP-YOLO provides a favorable trade-off between detection accuracy, model complexity, and edge inference efficiency, although its detection accuracy is slightly lower than that of the baseline. The proposed model therefore provides a lightweight candidate for further validation in continuous orchard monitoring applications.

Keywords: Korla fragrant pear; disease and pest detection; lightweight object detection; multi-organ dataset; edge deployment

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1. Introduction

China is one of the world's major pear-producing countries, and pear cultivation represents an important component of its fruit industry [1]. Korla fragrant pear is a

characteristic fruit cultivar of Xinjiang, China, with substantial regional economic value and an important role in orchard production and farmers' income generation [2]. However, its production is continuously affected by several diseases and pests, including pear fire blight, pear leaf spot disease, fruit infestation by *Grapholita molesta*, leaf infestation by pear gall midge, and other forms of leaf insect damage [3,4]. These biological stresses may affect different organs and developmental stages. Fire blight may damage flowers, shoots, leaves, and fruits, whereas leaf spot and chlorosis reduce the effective photosynthetic area and weaken canopy development. Insect feeding and gall-midge infestation may cause tissue injury, leaf deformation, and impaired vegetative growth, while *G. molesta* directly damages fruits and reduces their commercial quality [5,6]. Under natural orchard conditions, the visual appearance of these symptoms is further affected by illumination variation, canopy overlap, background clutter, and partial occlusion [7]. Failure to identify disease and pest symptoms in a timely manner may lead to fruit-quality deterioration, yield losses, and increased crop-protection costs [8]. Therefore, reliable and timely detection is important for orchard inspection, early intervention, and site-specific pest-management decisions.

In recent years, fruit-tree disease and pest detection has developed from conventional object detection methods, such as Faster R-CNN and SSD, toward YOLO-based end-to-end frameworks and Transformer-based feature-learning architectures [9–11]. Among these methods, the YOLO family has been widely adopted in agricultural vision because of its favorable balance between detection accuracy and inference efficiency [12]. Mohanty et al. demonstrated the feasibility of convolutional neural networks for image-based plant disease recognition across multiple crops and disease categories [13]. Ferentinos subsequently developed deep learning models for large-scale plant–disease classification, further confirming the effectiveness of CNN-based approaches in agricultural image analysis [14]. Lightweight architecture design has also received increasing attention. For example, Goyal et al. developed a computationally efficient CNN for citrus disease classification [15]. Improved YOLO models have also been applied to apple and grape disease detection by integrating optimized loss functions, attention mechanisms, re-parameterized modules, and multi-scale feature-fusion strategies [16–18]. These studies indicate that lightweight feature extraction, attention-based enhancement, and multi-scale representation are important directions for agricultural object detection.

Beyond standalone visual recognition, computer vision and machine learning are increasingly being integrated with environmental sensing, disease and pest forecasting, and agricultural decision-support systems. Such integrated approaches can support continuous crop-health monitoring, early risk identification, targeted intervention, and more efficient use of crop-protection inputs. Fanourakis et al. emphasized that environmental monitoring and pest- and disease-risk assessment can support preventive and climate-adaptive crop protection [19]. Their subsequent work further discussed the contribution of sensor-based monitoring, precision management, and crop-specific agronomic practices to production efficiency and the scalability of intelligent horticultural systems [20]. These developments suggest that disease and pest detectors should be evaluated not only by image-level accuracy but also by robustness, computational efficiency, and suitability for integration into practical monitoring workflows.

Research on pear disease and pest recognition has progressed from image classification toward object localization and multi-class detection. Yang et al. investigated the effects of image resolution and environmental conditions on pear disease recognition and showed that complex backgrounds and small lesion regions can substantially affect model performance [21]. Gu et al. applied transfer learning and deep convolutional networks to apple and pear disease recognition, demonstrating that pretrained models can alleviate limitations associated with insufficient training samples [22]. Hai et al. introduced an

attention-aggregation strategy for pear disease detection to improve feature fusion and robustness in complex environments [23]. Cao et al. developed a MobileNet-based model for Korla fragrant pear disease classification, demonstrating the potential of lightweight networks for resource-constrained agricultural applications [24]. With the development of object detection, Chen et al. investigated YOLO-based pear disease localization [25], Zheng et al. introduced a semi-supervised framework for pear leaf disease detection under limited annotation conditions [26], and Wang et al. combined lightweight feature extraction with YOLO-based detection to support pear disease and pest recognition and spraying-related applications [27].

Despite this progress, several limitations remain. First, many existing datasets contain a limited number of categories and are dominated by leaf images, whereas the joint detection of healthy, diseased, and pest-damaged targets across leaves, fruits, and flowers has received comparatively less attention. Second, small affected regions, visually similar symptoms, severe occlusion, and cluttered orchard backgrounds remain major sources of missed detections and class confusion. Third, constructing and annotating sufficiently diverse agricultural datasets remains labor-intensive. Recent research has explored generative semantic reconstruction and foundation-model-assisted workflows to convert visually heterogeneous agricultural images into more structured, annotation-ready vegetation representations [28]. Such approaches provide a complementary strategy for improving data preparation before detector training. Fourth, differences among indoor, field, and publicly available images may introduce acquisition-domain shifts. Failure-oriented studies in agricultural vision have shown that illumination, object scale, image quality, occlusion, and domain variation can substantially affect detection reliability [29]. Therefore, strict separation of training, validation, and test data, together with independent external evaluation, is necessary for a more reliable assessment of model generalization.

A further challenge concerns edge deployment. High-capacity detectors may achieve strong accuracy but often require substantial memory and computation. In contrast, orchard-monitoring devices must frequently operate with limited processing resources, restricted power budgets, and real-time requirements. Consequently, a model with the highest detection accuracy is not necessarily the most appropriate model for field deployment. Model selection should instead consider the combined effects of detection accuracy, parameter count, computational complexity, inference speed, and external-data performance.

To address these issues, this study develops KFP-YOLO, a lightweight multi-organ disease and pest detector based on YOLO26n. The main contributions are summarized as follows:

(1) Multi-organ dataset construction: The Korla Fragrant Pear Disease and Pest Dataset, named KFP-PDD, was constructed using self-collected field images, self-collected indoor images, and supplementary public images. It covers leaves, fruits, and flowers and contains 11 healthy, disease, and pest categories. The image-acquisition procedure and diagnostic confirmation are described in Section 2.1, while data screening, source separation, dataset division, and augmentation are presented in Section 2.2.

(2) Task-oriented lightweight architecture: KFP-YOLO introduces ADown into selected backbone downsampling stages, replaces selected C3k2 blocks with the C3-PD module, and incorporates CFA into multi-scale feature-fusion nodes. The contribution does not lie in claiming ADown, partial convolution, squeeze-and-excitation attention, or coordinate attention as new fundamental operations. Instead, it lies in their task-specific redesign, placement, and coordinated integration for multi-organ Korla fragrant pear disease and pest detection. The complete architecture and module implementations are described in Section 2.3.

(3) Accuracy–efficiency and external generalization evaluation: The model is evaluated using a unified experimental protocol and standard detection and deployment metrics, as described in Sections 2.4 and 2.5. Comparative results and class-confusion characteristics are reported in Section 3.1, module-level effects are examined in Section 3.2, and the accuracy–efficiency trade-off is analyzed using Pareto frontiers in Section 3.3. Performance on an independently prepared external test set is evaluated in Section 3.4, while representative detection results and feature-activation maps are presented in Section 3.5.

Overall, the objective of KFP-YOLO is not to maximize detection accuracy alone, but to obtain a more compact model while retaining competitive detection capability. The study therefore focuses on the trade-off among detection performance, computational complexity, and edge inference efficiency. Although the proposed model shows potential as a lightweight candidate for orchard monitoring, its performance in continuous, long-term, and geographically diverse field operations requires further validation.

2. Materials and Methods

2.1. Image Data Acquisition

The self-collected portion of the KFP-PDD dataset was acquired from six Korla fragrant pear orchards in the Aksu region of Xinjiang, China. The primary image-acquisition campaign was conducted from April to September 2025, and a supplementary campaign was carried out from April to June 2026 to increase the coverage of different environmental conditions and seasonal appearances of leaves, flowers, and fruits. The geographic coordinates, altitude, plot size, row spacing, plant spacing, and average tree height of the six orchards are summarized in Table 1. Approximately 30, 18, 24, 40, 110, and 70 trees were surveyed in Orchards 1–6, respectively, corresponding to approximately 292 trees in total. These values represent the approximate numbers of trees that were inspected, photographed, or used as sources of detached leaves, flowers, or fruits during the field surveys. Representative top-view images of the six orchards are presented in Figure 1. Images were captured using a Canon EOS 700D digital camera (Canon Inc., Tokyo, Japan) equipped with an 18–55 mm zoom lens and several high-resolution smartphones. A combined field and indoor imaging strategy was adopted to include both natural orchard conditions and controlled-background observations. Field images were acquired under natural illumination in the original orchard environment. Frontal, lateral, and top-view perspectives were used to record targets under different viewing angles, object scales, illumination conditions, canopy structures, degrees of occlusion, and background complexities. The acquisition distance and camera orientation were varied according to target location and visibility rather than being fixed at a single distance. Indoor images were mainly obtained from detached leaves, flowers, and fruits collected during orchard surveys. White A4 paper or black shading cloth was used as the background, and supplementary illumination was applied when necessary. Indoor imaging was used to provide clearer observations of lesion boundaries, tissue discoloration, insect morphology, feeding traces, and other fine-grained characteristics. These images were retained as a distinct acquisition domain rather than being treated as equivalent to natural field images. Sampling was primarily guided by the occurrence and visibility of disease and pest symptoms and by sample availability, rather than by a fixed sampling interval or a strictly phenological stage-matched design. Flowers were mainly photographed or collected during the flowering stage, leaves were sampled from the leaf-expansion stage to the mature-leaf stage, and fruits were sampled mainly from the young-fruit stage to the fruit-expansion stage. During each orchard survey, organs showing visible disease or pest symptoms, together with representative healthy organs, were recorded when available. Consequently, samples of the same organ from different

orchards were not necessarily collected at identical phenological stages. Because complete image-level phenological records were not available for all samples, detection performance at specific developmental stages was not evaluated separately in this study.

Table 1. Orchard information of the experimental sites.

Orchard No.	Location	Coordinates	Altitude (m)	Plot Size (m)	Row Spacing (m)	Plant Spacing (m)	Average Tree Height (m)
Orchard 1	Tarim University Experimental Field 1	40°32′24.98″ N, 81°17′55.00″ E	951.08	93 × 37	4.7	4.0	4.5
Orchard 2	Tarim University Experimental Field 2	40°32′29.57″ N, 81°17′32.27″ E	952.24	25 × 15	4.6	1.7	4.0
Orchard 3	Tarim University Experimental Field 3	40°32′31.11″ N, 81°17′30.46″ E	951.98	55 × 22	3.7	1.8	3.7
Orchard 4	Tarim University Experimental Field 4	40°32′42.60″ N, 81°17′17.01″ E	950.91	202 × 33	5.4	2.6	3.6
Orchard 5	First Division 12th Regiment, 7th Company Orchard Base	40°28′57.15″ N, 81°19′04.23″ E	1042.35	630 × 190	4.5	4.0	4.0
Orchard 6	First Division 9th Regiment Orchard Base	40°40′12.55″ N, 81°00′51.00″ E	995.70	780 × 130	4.3	3.8	4.3

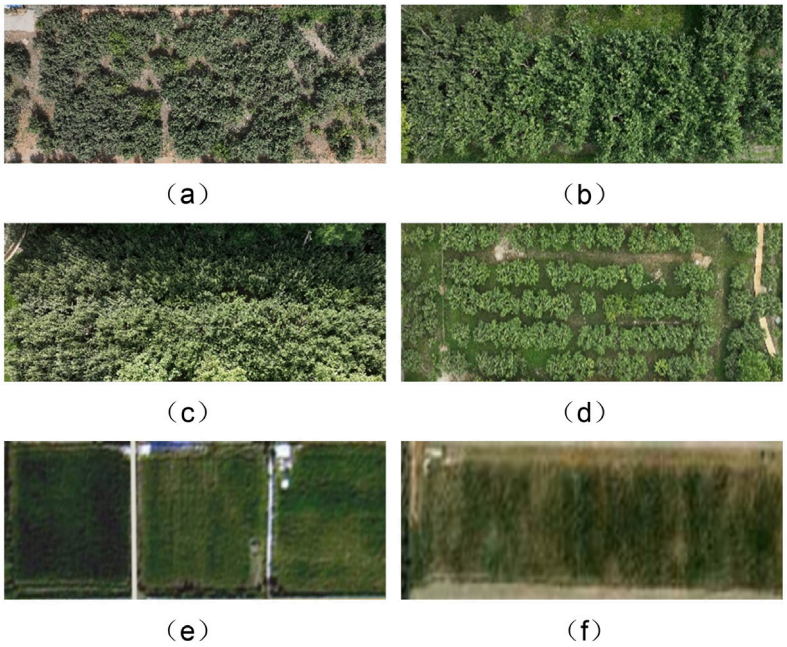


Figure 1. Top-view images of six pear orchards in the dataset: (**a**) Orchard 1; (**b**) Orchard 2; (**c**) Orchard 3; (**d**) Orchard 4; (**e**) Orchard 5; (**f**) Orchard 6.

The collected samples covered healthy organs and representative disease and pest categories, including pear fire blight, pear leaf spot disease, pear chlorosis disease, fruit infestation by *Grapholita molesta*, leaf infestation by pear gall midge, and other forms of leaf insect damage. The samples were organized according to three organ types—leaves, fruits, and flowers—because the occurrence and visual appearance of symptoms vary among organs. For example, leaf spot disease mainly affects leaves, and *G. molesta* primarily damages fruits, whereas fire blight may occur on leaves, flowers, and fruits [30]. The 11 categories included in KFP-PDD and their representative appearances are shown in Figure 2. Preliminary category identification was performed according to typical field

symptoms, lesion morphology, tissue discoloration, pest morphology, and characteristic feeding or infestation patterns. For representative infectious disease samples and samples with visually ambiguous symptoms, pathogen isolation and culture were conducted under laboratory conditions, followed by examination of colony characteristics and microscopic morphology. Pest categories were identified according to visible insect morphology and characteristic damage symptoms. The final category assignments were reviewed by plant-protection specialists from the College of Agronomy, Tarim University. Samples that could not be identified with sufficient confidence were excluded. Before annotation, all self-collected images were manually screened. Images were excluded when severe motion blur or defocus prevented reliable identification of the target boundary, substantial overexposure or underexposure obscured diagnostic characteristics, the image file was corrupted, or the image content was unrelated to the defined detection categories. Exact duplicate images were removed. Images showing the same target from nearly identical viewpoints and framing were treated as near-duplicates and were either removed or assigned to the same data group during dataset division, as described in Section 2.2.

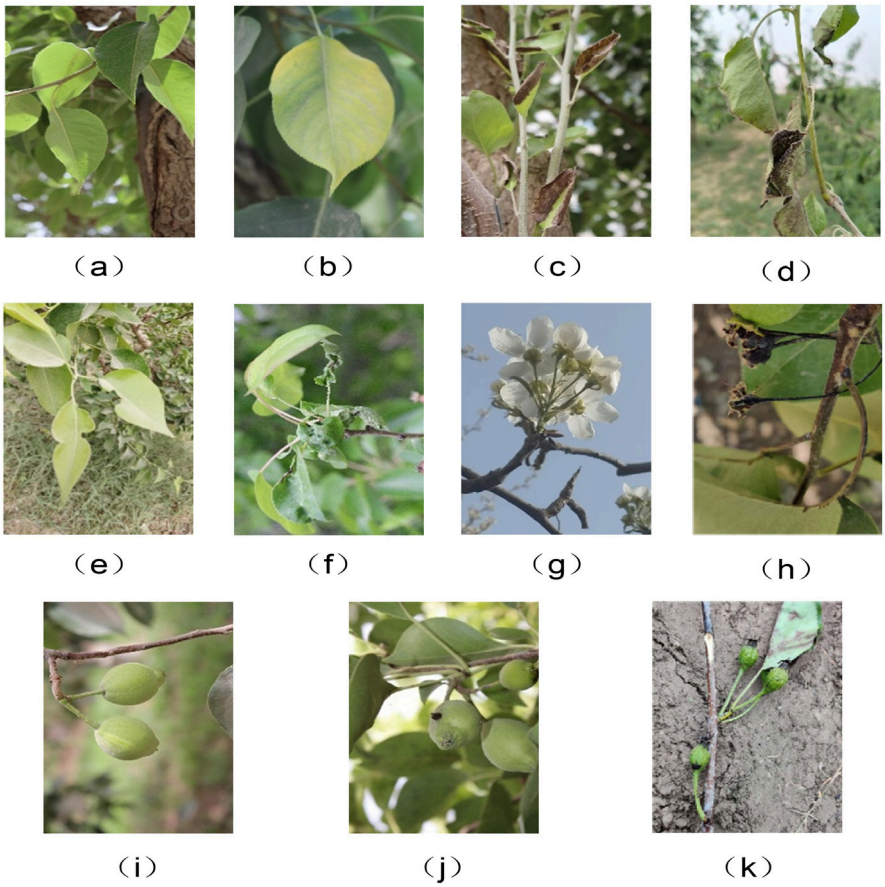


Figure 2. Representative sample images of pear leaf, flower, and fruit categories in the dataset: (**a**) Pear Leaf Healthy; (**b**) Pear Leaf Chlorosis Disease; (**c**) Pear Leaf Spot Disease; (**d**) Pear Leaf Fire Blight; (**e**) Pear Leaf Insect Damage; (**f**) Pear Leaf Infestation by Pear Gall Midge; (**g**) Pear Flower Healthy; (**h**) Pear Flower Fire Blight; (**i**) Pear Fruit Healthy; (**j**) Pear Fruit *Grapholita molesta* Infestation; and (**k**) Pear Fruit Fire Blight.

2.2. Dataset Construction

The original KFP-PDD dataset was constructed by integrating self-collected field images, self-collected indoor images, and publicly available images. The self-collected portion comprised 11,086 images, including 5223 field images and 5863 indoor images. The indoor images contained 5912 annotated instances. In addition, 3006 images from

the publicly available Kaggle Pear Leaf Disease dataset were incorporated to supplement the Pear Leaf Healthy and Pear Leaf Spot Disease categories. After multi-source data integration, the complete original KFP-PDD dataset contained 14,092 images and 22,735 annotated instances. The indoor images accounted for 41.61% of the original images, whereas their annotated instances accounted for 26.00% of all annotated instances.

Before annotation and dataset division, all self-collected and publicly available images were manually screened. Images were excluded if severe blur prevented clear identification of target boundaries, substantial overexposure or underexposure obscured disease and pest characteristics, image files were corrupted, or the image content was irrelevant to the detection task. Exact duplicate images were removed. Images showing the same organ or target from nearly identical viewpoints and framing were treated as near-duplicates and were either removed or retained within the same data group. Duplicate and near-duplicate checking was performed within the self-collected data, within the Kaggle data, and between the two data sources. All retained images were manually annotated using Labelme version 5.10.1. Each valid target was enclosed by a rectangular bounding box and assigned one of the 11 category labels. Each retained image contained at least one valid annotated object. The JSON annotation files generated by Labelme were subsequently converted into YOLO-format TXT files.

To prevent augmented derivatives of the same original image from appearing in different subsets, dataset division was completed before data augmentation. The 14,092 original images were divided into training, validation, and test subsets at an approximate ratio of 8:1:1 using class-stratified sampling. The training, validation, and test subsets contained 11,276, 1409, and 1407 images, respectively, with 18,172, 2276, and 2287 annotated instances. Images originating from the same continuous acquisition sequence, the same detached sample, or highly similar views of the same target were retained within the same subset whenever identifiable. The 3006 Kaggle images were assigned exclusively to the training subset, whereas the validation and test subsets contained only original self-collected images. Therefore, publicly available images did not directly contribute to model selection or performance evaluation on the internal KFP-PDD test set. In addition to the internal test subset, an independently prepared external test set was constructed to further evaluate model generalization. The external images were collected during separate acquisition sessions from April to July 2026 from the same six Korla fragrant pear orchards in the Aksu region of Xinjiang, China, as described in Table 1. All images were acquired under natural orchard conditions using a Canon EOS 700D digital camera (Canon Inc., Tokyo, Japan). The external test set contained 1052 images and 3377 annotated instances covering the same 11 categories as KFP-PDD. The external data were stored and screened independently from the development data. Although the external test set was collected from the same six orchards, it was acquired during separate acquisition sessions in 2026 and was independently screened against the development data. Exact duplicate images, images from the same continuous acquisition sequence, and identifiable near-duplicate views of the same target were excluded. Therefore, no exact duplicate or identifiable near-duplicate images were shared between the external test set and the training, validation, or internal test subsets. The external test set was not used for model training, validation, hyperparameter adjustment, or model selection. The class-wise distributions of the three internal subsets and the independent external test set before augmentation are presented in Table 2.

After dataset division, data augmentation was applied exclusively to the training subset. For indoor training images, the U^2-Net segmentation model was used to extract target regions and reduce redundant background interference [31]. Mosaic3×3 augmentation was subsequently applied to construct multi-object composite scenes and increase variations

in object scale and spatial layout [32]. Field training images were randomly divided into two approximately equal subsets. Random sandstorm noise was applied to one subset to simulate windy and dusty orchard conditions, whereas the other subset was processed using Fourier-domain transformation and Gaussian frequency-domain filtering to simulate variations in illumination, image sharpness, motion-related degradation, and sensor interference [33,34]. Consistent augmentation methods and parameter settings were used across categories under the same acquisition condition. The validation and internal test subsets were not augmented and retained their original image content. The independent external test set was also not subjected to data augmentation. During model training and evaluation, all images were resized to 640 × 640 pixels using the same preprocessing procedure. The augmentation procedures described above refer to offline dataset augmentation, whereas the online augmentation parameters used during model training are reported separately in Section 2.4. The external test set was evaluated using the same input resolution and preprocessing procedure as the internal test subset, and the corresponding results are presented in Section 3.4. Although the independent external test set provides additional evaluation under previously unseen natural orchard conditions, both the internal and external datasets mainly represent Korla fragrant pear production conditions in the Aksu region. Therefore, the present results do not demonstrate broad transferability across different pear cultivars, climatic regions, or production systems. Additional validation under different cultivar characteristics, climatic conditions, orchard structures, planting densities, and crop-management practices will be required before broad generalizability can be established.

Table 2. Class-wise distribution of the KFP-PDD dataset and independent external test set before data augmentation.

ID	Category	Train Images	Train Instances	Validation Images	Validation Instances	Test Images	Test Instances	External Test Images	External Test Instances
0	Pear Leaf Healthy	2235	3503	296	429	298	452	317	637
1	Pear Leaf Chlorosis Disease	1146	1749	144	217	143	214	237	525
2	Pear Leaf Spot Disease	4992	7338	671	980	618	929	459	760
3	Pear Leaf Fire Blight	420	457	51	62	54	60	22	42
4	Pear Leaf Insect Damage	595	654	61	63	84	86	80	111
5	Pear Leaf Infestation by Pear Gall Midge	346	448	42	46	58	58	25	55
6	Pear Fruit Healthy	1138	1382	142	172	126	157	247	479
7	Pear Fruit Infestation by *Grapholita molesta*	1181	1260	131	141	136	146	214	289
8	Pear Fruit Fire Blight	433	453	47	47	62	65	19	33
9	Pear Flower Healthy	145	462	23	73	22	55	37	313
10	Pear Flower Fire Blight	339	466	33	46	44	65	25	133
Total		11,276	18,172	1409	2276	1407	2287	1052	3377

Note: An image may contain objects belonging to more than one category. Therefore, the class-wise image counts are not mutually exclusive and do not sum to the total number of unique images in each subset. The External Test subset represents an independently collected natural orchard dataset and was not involved in model training or parameter selection.

2.3. Object Detection Model and Improvements

YOLO26 is an officially released real-time vision model family developed within the Ultralytics framework. It follows the continued evolution of the Ultralytics YOLO series, including YOLOv5, YOLOv8, and YOLO11, and is designed for efficient end-to-end inference and deployment. YOLO26 provides multiple model scales, among which YOLO26n is the nano-scale variant. In this study, YOLO26n was selected as the baseline because its compact architecture is suitable for resource-constrained edge deployment [35]. YOLO26 retains the

general backbone–neck–detection-head architecture of previous Ultralytics YOLO models and adopts a dual-branch training strategy. The one-to-many branch provides dense supervision during training, whereas the one-to-one branch is used for end-to-end inference. The one-to-one prediction branch directly generates detection results without requiring non-maximum suppression, thereby simplifying the inference process. In addition, YOLO26 removes Distribution Focal Loss from bounding-box regression and introduces Progressive Loss and Small-Target-Aware Label Assignment to improve supervision alignment and positive-sample assignment for small targets [35].

YOLO26n was selected as the baseline for three main reasons. First, it is an official lightweight variant within the Ultralytics model family and supports a unified workflow for training, validation, inference, and deployment. Second, its compact model scale and end-to-end inference mechanism are consistent with the real-time requirements of the Jetson AGX Orin platform. Third, its small-target-aware training strategy is relevant to the small lesions, pest-damaged regions, and partially occluded targets present in the KFP-PDD dataset. However, when directly applied to multi-organ Korla fragrant pear disease and pest detection, YOLO26n may still experience fine-grained information loss during repeated downsampling and redundant computation during feature extraction. These limitations are particularly relevant for small lesions, subtle texture differences, severe occlusion, and complex orchard backgrounds. Therefore, task-oriented lightweight redesign and feature enhancement were introduced to improve the balance between model complexity and detection performance. The detailed architectural modifications are presented in Sections 2.3.1–2.3.4.

2.3.1. KFP-YOLO Model Architecture

To adapt YOLO26n to multi-organ Korla fragrant pear disease and pest detection under resource-constrained edge deployment, selected downsampling, feature-extraction, and multi-scale feature-fusion stages were redesigned. The architectural modifications mainly address three task-related issues: fine-grained feature loss during repeated downsampling, redundant computation in intermediate feature extraction, and insufficient spatial-context modeling during multi-scale feature fusion.

As shown in Figure 3, the four stride-2 convolutional downsampling layers in the YOLO26n backbone were replaced with ADown modules. By combining average-pooling and max-pooling branches, ADown reduces spatial resolution while retaining complementary contextual and salient local information [36]. In the neck, the corresponding C3k2 feature-processing blocks were replaced with C3-PD modules. Each C3-PD module integrates partial convolution to reduce redundant channel-wise computation and squeeze-and-excitation attention to recalibrate informative feature responses [37,38]. In addition, the four conventional Concat nodes used for multi-scale feature fusion were replaced with CFA modules. Each CFA module first combines multi-scale feature inputs and then applies coordinate attention to encode directional positional information and recalibrate the fused features [39]. ADown, partial convolution, squeeze-and-excitation attention, and coordinate attention are established operations rather than newly proposed fundamental components. Accordingly, the methodological contribution of KFP-YOLO lies in the task-oriented construction of C3-PD and CFA, the placement of these modules within YOLO26n, and their coordinated integration for multi-organ pear disease and pest detection. The design objective is to reduce parameter count and computational complexity while retaining competitive detection performance, rather than to maximize detection accuracy alone. The overall architecture of KFP-YOLO is illustrated in Figure 3.

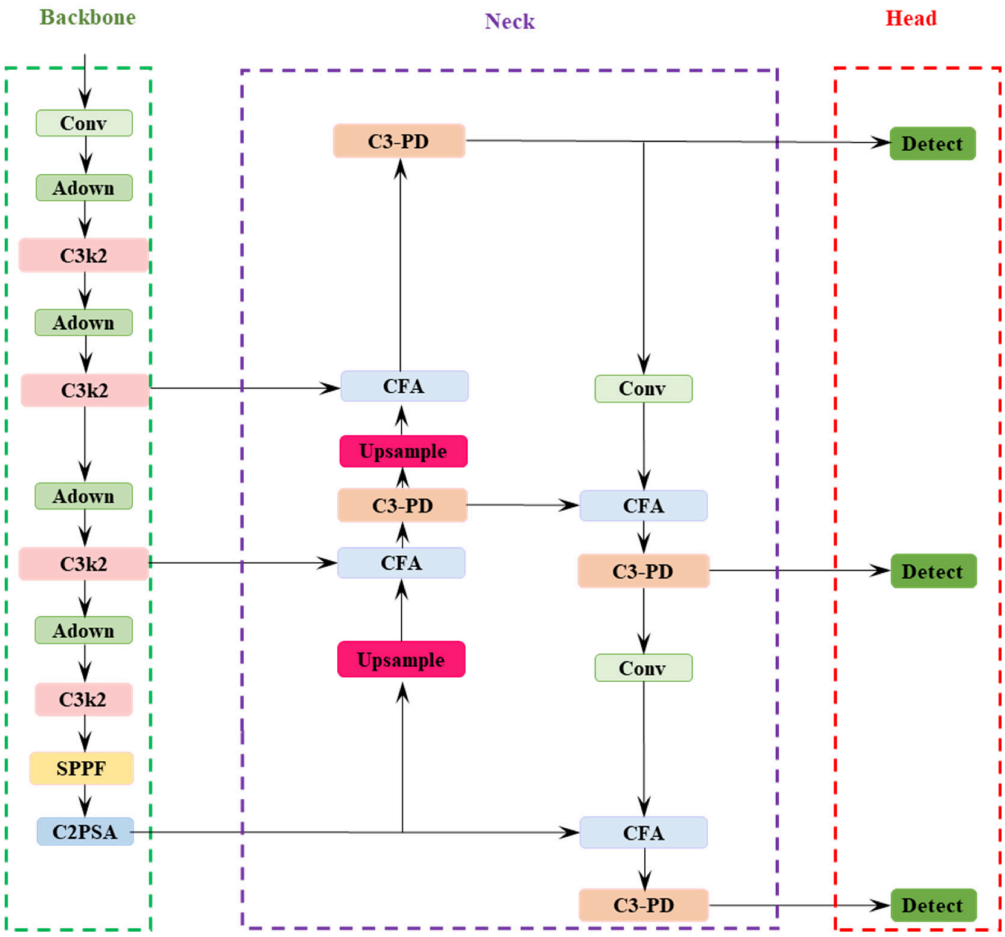


Figure 3. Overall architecture of KFP-YOLO and the placement of the ADown, C3-PD, and CFA modules. Different colors are used to distinguish the backbone, neck, detection head, and the introduced ADown, C3-PD, and CFA modules, as indicated by the labels in the figure.

2.3.2. Lightweight Downsampling Module ADown

Repeated downsampling in the backbone may weaken the representation of small lesions, pest-damaged regions, and fine-grained boundaries in multi-organ pear images. To reduce computational redundancy during spatial resolution reduction, an ADown variant module was introduced to replace selected downsampling convolution layers in the YOLO26n backbone [36]. ADown adopts a channel-splitting and dual-branch processing structure to extract complementary feature information through different operations.

As shown in Figure 4, the input feature map X is first processed by average pooling and is then divided into two sub-feature maps, X_1 and X_2, along the channel dimension, as expressed in Equation (1). In the first branch, X_1 is processed by a convolution operation to obtain Y_1, as defined in Equation (2). In the second branch, X_2 is first processed by max pooling and is subsequently transformed by a convolution operation to obtain Y_2, as expressed in Equation (3). Finally, Y_1 and Y_2 are concatenated along the channel dimension to generate the output feature map Y, as defined in Equation (4). The computational process is expressed as follows:

$$X_1, X_2 = \mathrm{Split}\left(\mathrm{AvgPool2d}_{2\times 2, s=2}(X)\right) \tag{1}$$

$$Y_1 = \mathrm{Conv}_{3\times 3, s=1}(X_1) \tag{2}$$

$$Y_2 = \mathrm{Conv}_{1\times 1, s=1}(\mathrm{MaxPool2d}_{2\times 2, s=1}(X_2)) \tag{3}$$

$$Y = \mathrm{Concat}(Y_1, Y_2) \tag{4}$$

where X denotes the input feature map; X_1 and X_2 denote the two sub-feature maps obtained after average pooling and channel-wise splitting; Y_1 and Y_2 represent the outputs of the two processing branches; and Y denotes the final output obtained through channel-wise concatenation.

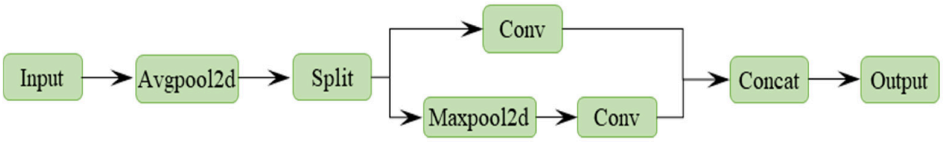


Figure 4. Structure of the ADown lightweight downsampling module.

In this structure, the average-pooling operation first aggregates local feature information, after which the two branches perform different feature transformations. The first branch applies convolution to extract spatial features, whereas the second branch combines max pooling and convolution to retain relatively strong local responses and perform channel transformation. The concatenation of the two branch outputs integrates the complementary information extracted by the two processing paths. For Korla fragrant pear disease and pest detection, the ADown module is intended to reduce computational redundancy during downsampling while retaining useful lesion- and pest-related feature responses. Its effects on model complexity, inference speed, and detection performance are quantitatively evaluated through the ablation experiments in Section 3.2.

2.3.3. Lightweight Feature Extraction Module C3-PD

To reduce redundant computation while preserving the representation of fine-grained disease and pest features, the C3-PD module was designed to replace selected C3k2 feature-processing blocks in the YOLO26n network. As shown in Figure 5, the C3-PD module adopts a dual-branch structure. After the input feature map enters the module, two parallel convolution operations are first used for channel adjustment and preliminary feature mapping. One branch is used to preserve relatively shallow feature information, whereas the other branch performs deeper feature extraction through a sequence of PDBlocks. The outputs of the two branches are then concatenated and further transformed by convolution to obtain the final output feature map.

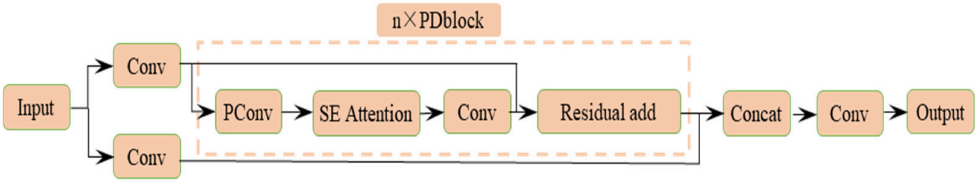


Figure 5. Structure of the C3-PD feature extraction module.

The PDBlock is the core component of the C3-PD module. It integrates partial convolution and squeeze-and-excitation attention to reduce redundant channel-wise computation and enhance the response to informative disease and pest features [37,38]. Partial convolution performs convolutional computation on only a subset of the input channels, while the remaining channels are retained or directly propagated. This strategy reduces computational cost while preserving part of the original feature information. The processed and unprocessed features are subsequently fused to form the output of the partial convolution operation.

The feature map generated by partial convolution is then passed through the squeeze-and-excitation attention mechanism. Global average pooling is first used to aggregate channel-wise information. The resulting channel descriptor is then transformed through two fully connected layers and a nonlinear activation function to generate channel weights. These weights are applied to the input feature map to strengthen informative channels and suppress less relevant responses. A convolution operation is subsequently used for further feature transformation.

To maintain stable information propagation and alleviate feature degradation in deeper networks, a residual connection is introduced within the PDBlock. The transformed feature is added to the original input feature, and the resulting output is passed to the subsequent layer. The computational process is expressed in Equations (5) and (6).

$$Y = \mathrm{Conv}_{1\times1,s=1}(\mathrm{SE}(\mathrm{PConv}(X))) \tag{5}$$

$$Z = X + Y \tag{6}$$

By combining partial convolution, channel attention, and residual feature propagation, the C3-PD module reduces redundant computation while retaining useful shallow and deep feature information. For Korla fragrant pear disease and pest detection, this module is intended to improve the representation of subtle lesions, feeding traces, and small pest-related regions under complex backgrounds. Its actual influence on parameter count, computational complexity, inference speed, and detection performance is evaluated in the ablation experiments presented in Section 3.2.

2.3.4. CFA Feature Enhancement and Multi-Scale Fusion Module

To improve multi-scale feature fusion and enhance the spatial representation of disease and pest targets, the CFA module was introduced to replace selected Concat operations in the YOLO26n neck. As shown in Figure 6, CFA first concatenates feature maps from different scales along the channel dimension to obtain a fused feature representation. Coordinate attention is then applied to encode directional spatial information and recalibrate the fused features [39].

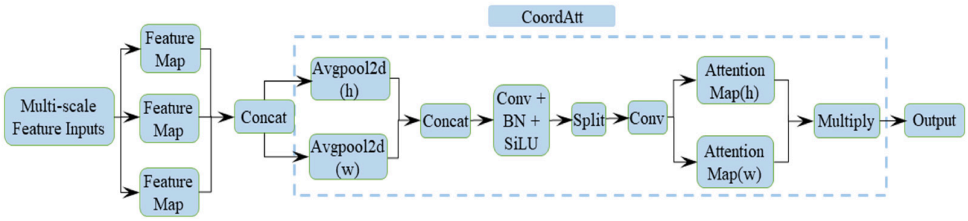


Figure 6. Structure of the CFA feature fusion module.

Specifically, the multi-scale input feature maps are first concatenated along the channel dimension. The fused feature map is then separately pooled along the horizontal and vertical directions to generate two directional feature descriptors. Unlike conventional global average pooling, which compresses spatial information in both directions simultaneously, the two directional pooling operations preserve positional information along one spatial dimension while aggregating contextual information along the other.

The horizontal and vertical feature descriptors are subsequently concatenated and passed through a shared convolution operation, batch normalization, and a SiLU activation function for feature transformation and channel compression. The resulting intermediate feature is then divided into horizontal and vertical components according to the spatial dimension. Two independent convolution operations are applied to restore the origi-

nal channel dimensions, after which Sigmoid activation functions are used to generate the corresponding horizontal and vertical attention weights. The generated attention weights are applied to the fused input feature map through element-wise multiplication. In this way, the module recalibrates feature responses according to both channel dependencies and directional positional information. The computational process is expressed in Equations (7)–(11).

$$X' = \mathrm{Concat}\left(\mathrm{AvgPool2d}_{H\times 1, s=1}(X), \mathrm{AvgPool2d}_{1\times W, s=1}(X)\right) \tag{7}$$

$$Y = \delta\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 1, s=1}(X')\right)\right) \tag{8}$$

$$a_h = \sigma\left(\mathrm{Conv}^{\mathrm{h}}_{1\times 1, s=1}\left(Y^h\right)\right) \tag{9}$$

$$a_w = \sigma\left(\mathrm{Conv}^{\mathrm{w}}_{1\times 1, s=1}(Y^w)\right) \tag{10}$$

$$Z = X \times a_h \times a_w \tag{11}$$

where X denotes the fused input feature map; X' denotes the concatenated directional feature descriptor obtained after horizontal and vertical pooling; Y denotes the intermediate feature generated through shared convolution, batch normalization, and SiLU activation; Y^h and Y^w denote the horizontal and vertical components split from Y, respectively; a_h and a_w denote the horizontal and vertical attention weights, respectively; and Z denotes the final output feature map after directional attention weighting.

By integrating multi-scale feature concatenation with coordinate attention, CFA retains the original feature-fusion function of Concat while additionally introducing directional spatial encoding and adaptive feature weighting. For Korla fragrant pear disease and pest detection, this module is intended to enhance the representation of spatially distributed lesion and pest features and reduce irrelevant responses from complex orchard backgrounds. Its effects on detection accuracy, parameter count, computational complexity, and inference efficiency are quantitatively evaluated through the ablation experiments in Section 3.2.

2.4. Experimental Environment and Parameters

The hardware and software configurations used for model training and edge deployment are summarized in Table 3. Model training was performed on a high-performance workstation, whereas edge inference was conducted on an NVIDIA Jetson AGX Orin Developer Kit (NVIDIA Corporation, Santa Clara, CA, USA) to evaluate the deployment efficiency of the proposed model in an edge-computing environment.

Table 3. Configuration of model training and deployment environments.

Item	Training Environment (PC)	Deployment Environment (Jetson AGX Orin)
Operating system	Windows 11	Ubuntu 22.04
CPU	Intel i9-13900KF	12-core Arm Cortex-A78AE
GPU	NVIDIA RTX 4080	Ampere GPU (2048 CUDA cores)
Memory	32 GB	64 GB LPDDR5
Python	3.12	3.10
PyTorch	PyTorch 2.5.1	PyTorch 2.1
CUDA	12.1	12.2
Ultralytics	8.4.0	—
THOP	2.0.18	—
ONNX	1.16.0	—
TensorRT	—	10.0.1

The training environment consisted of a workstation running Windows 11, equipped with an Intel Core i9-13900KF CPU, an NVIDIA GeForce RTX 4080 GPU, and 32 GB of RAM. The software environment included Python 3.12, PyTorch 2.5.1, CUDA 12.1, and Ultralytics 8.4.0. THOP 2.0.18 was employed to calculate the model parameters and computational complexity, while ONNX 1.16.0 was used to export the trained PyTorch models for subsequent deployment. All models were trained using an input resolution of 640 × 640 pixels for 300 epochs with a training batch size of 32. The stochastic gradient descent (SGD) optimizer was adopted, with an initial learning rate of 0.01 and a final learning-rate factor of 0.01. To evaluate the statistical stability of the experimental results, each comparison and ablation model was independently trained using three random seeds: 0, 42, and 134. The same set of random seeds was used for all models, and the detection metrics are reported as the mean ± standard deviation of the three independent runs. To ensure a fair comparison, all baseline, comparison, and ablation models were trained using identical dataset partitions, training epochs, optimizer settings, input resolution, and data-augmentation parameters.

In addition to the offline data augmentation described in Section 2.2, online data augmentation provided by the Ultralytics framework was enabled during model training. The augmentation parameters included HSV augmentation (hsv_h = 0.015, hsv_s = 0.7, and hsv_v = 0.4), random translation (translate = 0.1), random scaling (scale = 0.5), horizontal flipping (fliplr = 0.5), and Mosaic augmentation (mosaic = 1.0). These augmentation operations were applied only to the training set.

For edge deployment, the trained PyTorch models were first exported to ONNX format on the PC platform and subsequently converted into TensorRT inference engines on the Jetson AGX Orin platform. The deployment environment consisted of Ubuntu 22.04, Python 3.10, PyTorch 2.1, CUDA 12.2, and TensorRT 10.0.1. During inference benchmarking, the Jetson AGX Orin was configured in MAXN power mode, and the jetson_clocks utility was enabled to lock the CPU, GPU, and memory frequencies at their maximum supported values. The TensorRT engines were built and executed in FP16 precision mode. The inference batch size was set to 1, with an input resolution of 640 × 640 pixels.

Before formal benchmarking, 100 warm-up iterations were performed to complete runtime initialization and stabilize inference latency. These warm-up iterations were excluded from the latency and FPS calculations. A total of 1000 test images were then processed consecutively on the Jetson AGX Orin platform. The reported end-to-end latency included image resizing, normalization, TensorRT inference, output decoding, model-specific post-processing, and bounding-box coordinate mapping, while image loading, result visualization, and result saving were excluded from the timing process. The average processing latency per image was calculated over the complete test sequence, and the corresponding FPS was obtained from the average end-to-end processing time. Unless otherwise specified, all comparative models were evaluated under the same deployment environment and inference protocol.

2.5. Evaluation Metrics

Precision (P), Recall (R), mean Average Precision at an IoU threshold of 0.5 (mAP@0.5), and mean Average Precision averaged over IoU thresholds from 0.5 to 0.95 (mAP@0.5:0.95) were adopted as the primary detection metrics. Precision measures the proportion of correctly predicted positive objects among all predicted positive objects, whereas Recall evaluates the proportion of ground-truth objects successfully detected by the model. mAP@0.5 evaluates detection performance under a relatively relaxed localization criterion, while mAP@0.5:0.95 provides a more stringent assessment by averaging the AP values over multiple IoU thresholds. As described in Section 2.4, these detection metrics are reported

as the mean ± standard deviation of three independent runs using random seeds 0, 42, and 134.

Model Size (MB), Params (M), and GFLOPs were used to evaluate model complexity. Model Size denotes the storage size of the trained PyTorch weight file and reflects the model storage requirement rather than runtime memory consumption. Params represents the total number of learnable parameters in the network. GFLOPs denotes the number of floating-point operations required for one forward pass with an input resolution of 640 × 640 pixels. In this study, Params and GFLOPs were calculated from the PyTorch-format models using THOP 2.0.18.

Frames Per Second (FPS) was used to evaluate inference efficiency on the NVIDIA Jetson AGX Orin platform. After the PyTorch models were exported through ONNX and converted into FP16 TensorRT engines, FPS was calculated from the average end-to-end processing time of 1000 consecutive test images. The timing process included image resizing, normalization, TensorRT inference, output decoding, model-specific post-processing, and bounding-box coordinate mapping, while image loading, result visualization, and result saving were excluded. Therefore, the reported FPS characterizes the inference efficiency of each model under the specified Jetson AGX Orin benchmarking protocol.

3. Experimental Results and Analysis

3.1. Comparative Experiments

A series of comparative experiments was conducted using representative lightweight and medium-scale object detection models, including YOLOv5n, YOLOv8n, YOLO11n, YOLO12n, YOLOv13n, YOLOv13s, YOLO26n, YOLO26s, and KFP-YOLO. All models were trained using the same KFP-PDD dataset partitions, input resolution, training epochs, optimizer settings, and data-augmentation parameters. The detection metrics are reported as the mean ± standard deviation of three independent runs using random seeds 0, 42, and 134. Model size, parameter count, computational complexity, and inference speed on the Jetson AGX Orin platform are also reported in Table 4.

Table 4. Performance comparison of different object detection models.

Model	Size (MB)	Params (M)	GFLOPs	FPS	mAP@0.5	mAP@0.5:0.95	Precision	Recall
YOLOv5n	5.04	2.51	7.19	321.85	0.9252 ± 0.0037	0.6970 ± 0.0053	0.9089 ± 0.0037	0.8651 ± 0.0071
YOLOv8n	5.98	3.01	8.20	311.78	0.9420 ± 0.0023	0.7516 ± 0.0038	0.9230 ± 0.0045	0.8932 ± 0.0054
YOLO11n	5.23	2.59	6.45	266.95	0.9435 ± 0.0021	0.7550 ± 0.0049	0.9236 ± 0.0019	0.8860 ± 0.0051
YOLO12n	5.28	2.57	6.49	143.02	0.9478 ± 0.0024	0.7808 ± 0.0027	0.9372 ± 0.0022	0.8980 ± 0.0044
YOLOv13n	5.17	2.46	6.30	109.32	0.9508 ± 0.0013	0.8065 ± 0.0038	0.9366 ± 0.0032	0.9048 ± 0.0052
YOLOv13s	18.76	9.55	22.30	73.08	0.9612 ± 0.0017	0.8715 ± 0.0046	0.9508 ± 0.0029	0.9132 ± 0.0020
YOLO26n	5.16	2.51	5.79	259.79	0.9458 ± 0.0022	0.7579 ± 0.0044	0.9268 ± 0.0033	0.8984 ± 0.0051
YOLO26s	19.40	9.96	22.54	167.67	0.9628 ± 0.0021	0.8362 ± 0.0035	0.9401 ± 0.0029	0.9152 ± 0.0027
KFP-YOLO	4.05	1.93	4.42	278.97	0.9429 ± 0.0015	0.7434 ± 0.0043	0.9261 ± 0.0031	0.8912 ± 0.0047

Among the compared models, YOLO26s achieved the highest mAP@0.5 of 0.9628, whereas YOLOv13s obtained the highest mAP@0.5:0.95 of 0.8715. However, these two medium-scale models contained 9.96 M and 9.55 M parameters and required 22.54 and 22.30 GFLOPs, respectively, resulting in substantially higher computational costs. YOLOv5n achieved the highest inference speed of 321.85 FPS, but its mAP@0.5 and mAP@0.5:0.95 were relatively lower, at 0.9252 and 0.6970. The YOLO26n baseline achieved an mAP@0.5 of 0.9458, an mAP@0.5:0.95 of 0.7579, a Precision of 0.9268, and a Recall of 0.8984, with 2.51 M parameters, 5.79 GFLOPs, and an inference speed of 259.79 FPS. KFP-YOLO achieved an mAP@0.5 of 0.9429, an mAP@0.5:0.95 of 0.7434, a Precision of 0.9261, and a Recall of

0.8912. Compared with YOLO26n, its mAP@0.5 and mAP@0.5:0.95 decreased by 0.29 and 1.45 percentage points, respectively, whereas the parameter count and GFLOPs decreased from 2.51 M and 5.79 to 1.93 M and 4.42, corresponding to reductions of 23.1% and 23.7%. Its model size decreased from 5.16 MB to 4.05 MB, and its inference speed increased by 7.4% to 278.97 FPS. Therefore, KFP-YOLO does not achieve the highest detection accuracy, but provides a favorable trade-off among detection performance, model complexity, and edge inference efficiency.

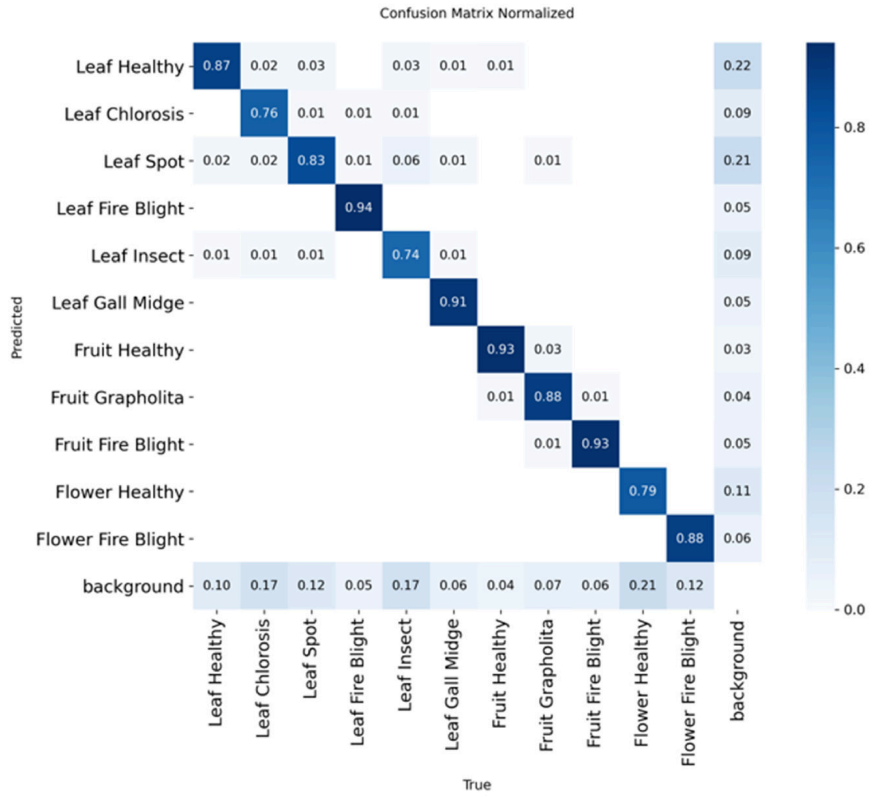
To provide a more detailed category-level evaluation of the final model, the class-wise Precision, Recall, AP@0.5, AP@0.5:0.95, and number of test instances for all 11 categories of KFP-YOLO are presented in Table 5. Consistent with the overall evaluation protocol, the class-wise detection metrics are reported as the mean ± standard deviation of three independent runs using random seeds 0, 42, and 134. As shown in Table 5, KFP-YOLO achieved relatively stable performance across the 11 categories. Pear Fruit Fire Blight obtained the highest AP@0.5 of 0.9672 and the highest Recall of 0.9397, while Pear Fruit Healthy achieved the highest Precision of 0.9644. Pear Flower Fire Blight and Pear Gall Midge infestation showed comparatively high AP@0.5:0.95 values of 0.8705 and 0.8518, respectively, indicating relatively robust localization performance under stricter IoU thresholds. In contrast, Pear Leaf Chlorosis Disease, Pear Leaf Insect Damage, and Pear Flower Healthy exhibited relatively lower Recall or AP values, suggesting that these categories remain more challenging to detect. Overall, the consistently higher AP@0.5 than AP@0.5:0.95 across all categories indicates that accurate localization under stricter overlap criteria remains a major source of performance loss.

Table 5. Class-wise detection performance of KFP-YOLO on the internal KFP-PDD test set.

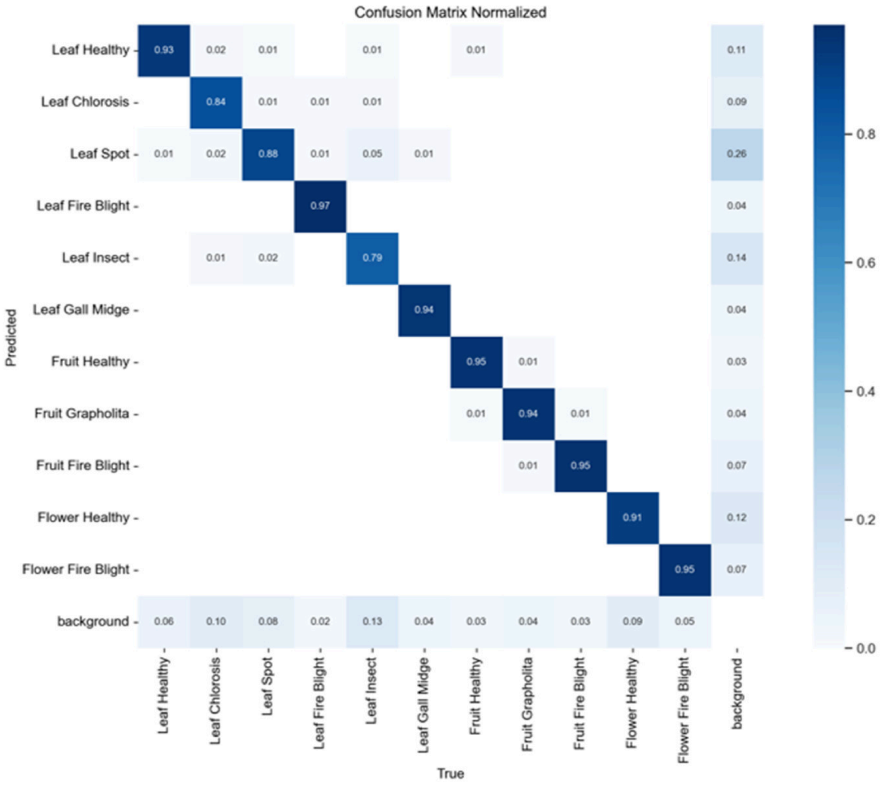
ID	Category	Test Instances	AP@0.5	AP@0.5:0.95	Precision	Recall
0	Pear Leaf Healthy	452	0.9456 ± 0.0016	0.7077 ± 0.0022	0.9321 ± 0.0060	0.9023 ± 0.0066
1	Pear Leaf Chlorosis Disease	214	0.8957 ± 0.0012	0.6458 ± 0.0021	0.9264 ± 0.0058	0.8382 ± 0.0071
2	Pear Leaf Spot Disease	929	0.9328 ± 0.0012	0.6888 ± 0.0021	0.9177 ± 0.0056	0.8762 ± 0.0070
3	Pear Leaf Fire Blight	60	0.9603 ± 0.0021	0.7197 ± 0.0031	0.9382 ± 0.0070	0.9392 ± 0.0094
4	Pear Leaf Insect Damage	86	0.9043 ± 0.0010	0.7073 ± 0.0051	0.8889 ± 0.0066	0.8229 ± 0.0044
5	Pear Leaf Infestation by Pear Gall Midge	58	0.9592 ± 0.0011	0.8518 ± 0.0052	0.9304 ± 0.0071	0.9299 ± 0.0086
6	Pear Fruit Healthy	157	0.9652 ± 0.0014	0.7143 ± 0.0044	0.9644 ± 0.0065	0.9135 ± 0.0058
7	Pear Fruit Infestation by *Grapholita molesta*	146	0.9610 ± 0.0009	0.7672 ± 0.0031	0.9413 ± 0.0093	0.9166 ± 0.0099
8	Pear Fruit Fire Blight	65	0.9672 ± 0.0006	0.8407 ± 0.0040	0.9017 ± 0.0054	0.9397 ± 0.0085
9	Pear Flower Healthy	55	0.9227 ± 0.0011	0.6549 ± 0.0045	0.8880 ± 0.0077	0.8275 ± 0.0068
10	Pear Flower Fire Blight	65	0.9557 ± 0.0010	0.8705 ± 0.0031	0.9308 ± 0.0061	0.9116 ± 0.0073

To further examine class-wise detection performance, normalized confusion matrices were generated for YOLOv5n, YOLOv13s, YOLO26n, and KFP-YOLO, as shown in Figure 7. The columns represent the ground-truth categories, whereas the rows represent the predicted categories. Values on the main diagonal indicate correctly classified targets, while off-diagonal values indicate confusion between categories. Entries in the background row mainly correspond to missed detections, whereas entries in the background column reflect false-positive predictions. The responses of the four models were concentrated mainly along the main diagonal, indicating that most healthy, disease, and pest categories were correctly distinguished. YOLOv13s showed relatively stronger class discrimination, consistent with its higher overall detection metrics, whereas YOLOv5n exhibited more evident inter-class confusion and background-related errors. The distributions of YOLO26n and KFP-YOLO were broadly similar, although residual errors remained for visually similar

categories and for small, occluded, or weakly expressed targets. These results indicate that visual similarity, limited target scale, occlusion, and complex orchard backgrounds remain important sources of detection error.

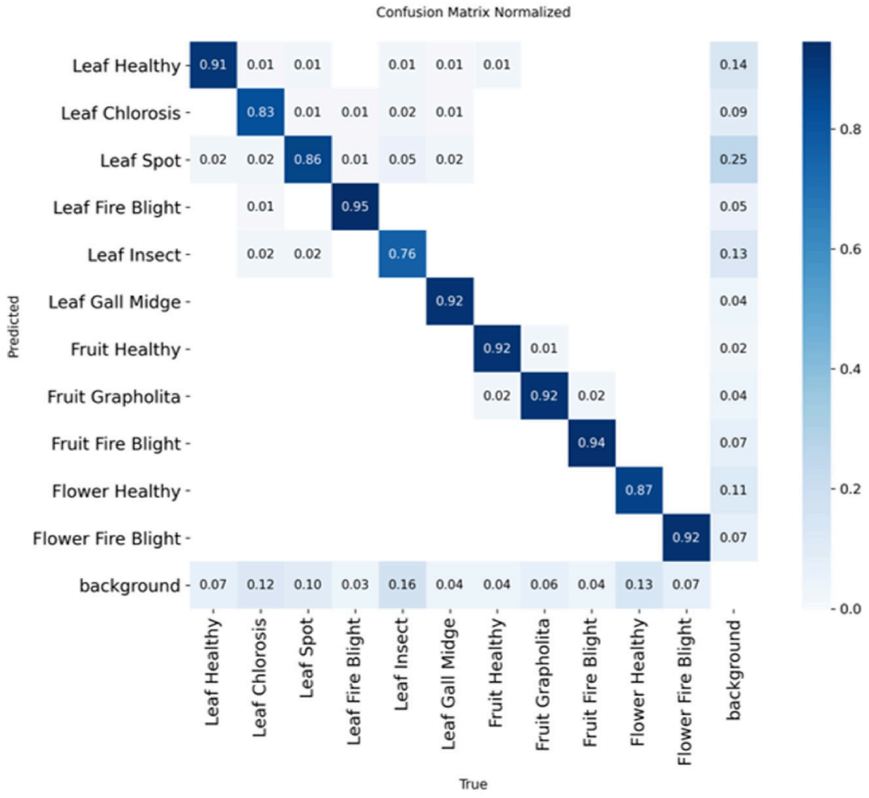


(**a**)

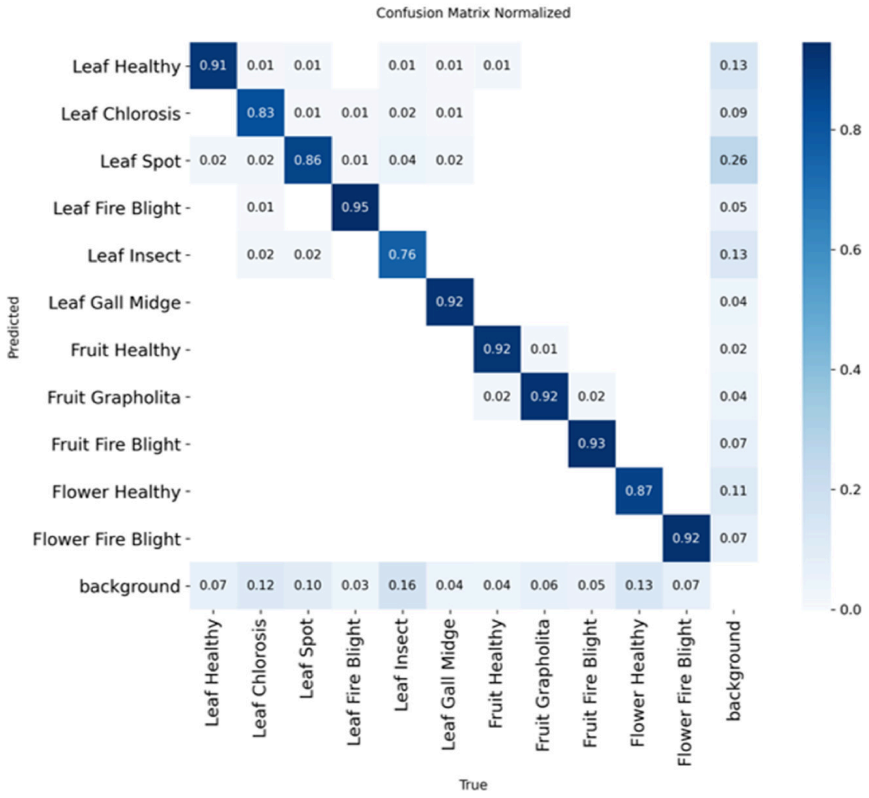


(**b**)

Figure 7. *Cont*.



(**c**)



(**d**)

Figure 7. Normalized confusion matrices of representative models on the KFP-PDD test set: (**a**) YOLOv5n; (**b**) YOLOv13s; (**c**) YOLO26n; (**d**) KFP-YOLO.

3.2. Ablation Experiments

Stepwise ablation experiments were conducted on the YOLO26n baseline to evaluate the effects of ADown, C3-PD, and CFA, as shown in Table 6. All detection metrics are reported as the mean ± standard deviation of three independent runs. Introducing ADown reduced the parameter count from 2.51 M to 2.27 M and GFLOPs from 5.79 to 5.09, while increasing the inference speed from 259.79 to 272.43 FPS. Its mAP@0.5 and mAP@0.5:0.95 decreased slightly by 0.29 and 0.68 percentage points, respectively. C3-PD produced the largest speed improvement among the individual modules, increasing FPS to 304.86 and reducing the parameter count to 2.13 M. However, its mAP@0.5:0.95 decreased from 0.7579 to 0.7134, indicating that stronger feature compression reduced detection performance under stricter localization criteria. CFA achieved an mAP@0.5 of 0.9482, which was 0.24 percentage points higher than that of the baseline, while reducing the parameter count and GFLOPs to 2.37 M and 5.35, respectively. However, its mAP@0.5:0.95 and Recall were slightly lower than those of YOLO26n, indicating that CFA did not improve all detection metrics simultaneously.

Table 6. Ablation experiment results.

Model	Size (MB)	Params (M)	GFLOPs	FPS	mAP@0.5	mAP@0.5:0.95	Precision	Recall
YOLO26n	5.16	2.51	5.79	259.79	0.9458 ± 0.0022	0.7579 ± 0.0044	0.9268 ± 0.0033	0.8984 ± 0.0051
+ADown	4.68	2.27	5.09	272.43	0.9429 ± 0.0023	0.7511 ± 0.0042	0.9264 ± 0.0031	0.8971 ± 0.0053
+C3-PD	4.37	2.13	5.12	304.86	0.9390 ± 0.0025	0.7134 ± 0.0048	0.9186 ± 0.0035	0.8769 ± 0.0055
+CFA	4.56	2.37	5.35	267.94	0.9482 ± 0.0011	0.7410 ± 0.0031	0.9270 ± 0.0030	0.8918 ± 0.0042
+ADown+C3-PD	3.92	1.89	4.41	291.62	0.9402 ± 0.0024	0.7352 ± 0.0045	0.9179 ± 0.0034	0.8888 ± 0.0054
+ADown+CFA	4.61	2.22	5.03	270.73	0.9527 ± 0.0010	0.7582 ± 0.0030	0.9268 ± 0.0029	0.9011 ± 0.0040
+C3-PD+CFA	4.47	2.17	5.13	286.41	0.9415 ± 0.0013	0.7168 ± 0.0037	0.9367 ± 0.0032	0.8914 ± 0.0043
KFP-YOLO	4.05	1.93	4.42	278.97	0.9429 ± 0.0015	0.7434 ± 0.0043	0.9261 ± 0.0031	0.8912 ± 0.0047

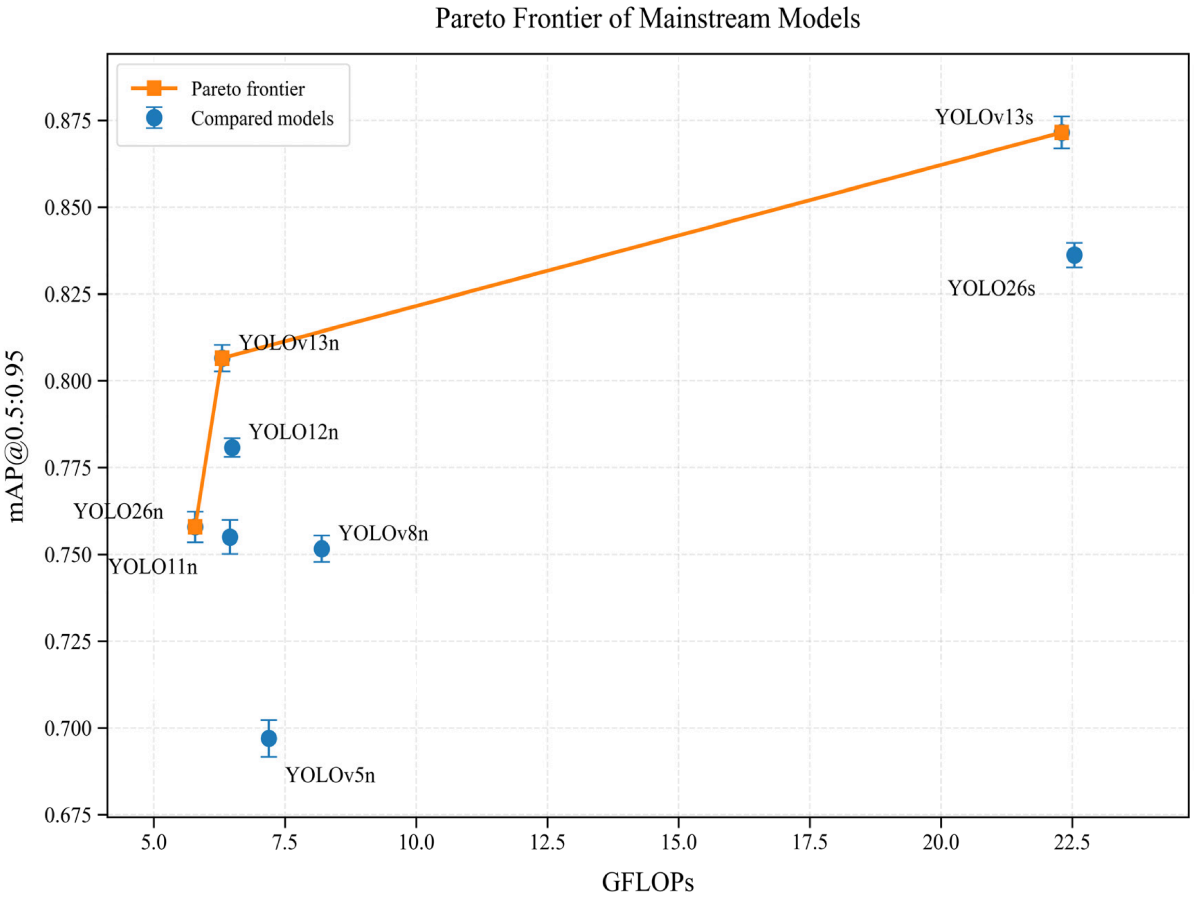
The combined-module experiments showed that the effects of the three modules were not simply additive. The ADown+C3-PD combination reduced the parameter count and GFLOPs to 1.89 M and 4.41, respectively, but its mAP@0.5 and mAP@0.5:0.95 decreased to 0.9402 and 0.7352. The ADown+CFA combination achieved the highest detection performance among the ablation configurations, with an mAP@0.5 of 0.9527, an mAP@0.5:0.95 of 0.7582, and a Recall of 0.9011, while maintaining 2.22 M parameters and 5.03 GFLOPs. The C3-PD+CFA combination reached the highest Precision of 0.9367 and an inference speed of 286.41 FPS, but its mAP@0.5:0.95 remained relatively low at 0.7168. These results indicate that different module combinations produce distinct trade-offs between feature representation, model compression, and inference efficiency.

After integrating all three modules, KFP-YOLO contained 1.93 M parameters and required 4.42 GFLOPs, representing reductions of 23.1% and 23.7%, respectively, compared with YOLO26n. Its inference speed increased by 7.4% to 278.97 FPS. Meanwhile, KFP-YOLO achieved an mAP@0.5 of 0.9429, an mAP@0.5:0.95 of 0.7434, a Precision of 0.9261, and a Recall of 0.8912, corresponding to decreases of 0.29, 1.45, 0.07, and 0.72 percentage points relative to the baseline. Although the ADown+CFA configuration achieved higher detection accuracy, KFP-YOLO further reduced the parameter count and GFLOPs by approximately 13.1% and 12.1%, respectively, and increased inference speed by approximately 3.0% compared with ADown+CFA. Therefore, the complete KFP-YOLO architecture was selected because the objective of this study was to obtain a more compact model with competitive detection performance and improved edge inference efficiency, rather than to select the configuration with the highest detection accuracy alone.

3.3. Accuracy–Efficiency Trade-Off Analysis

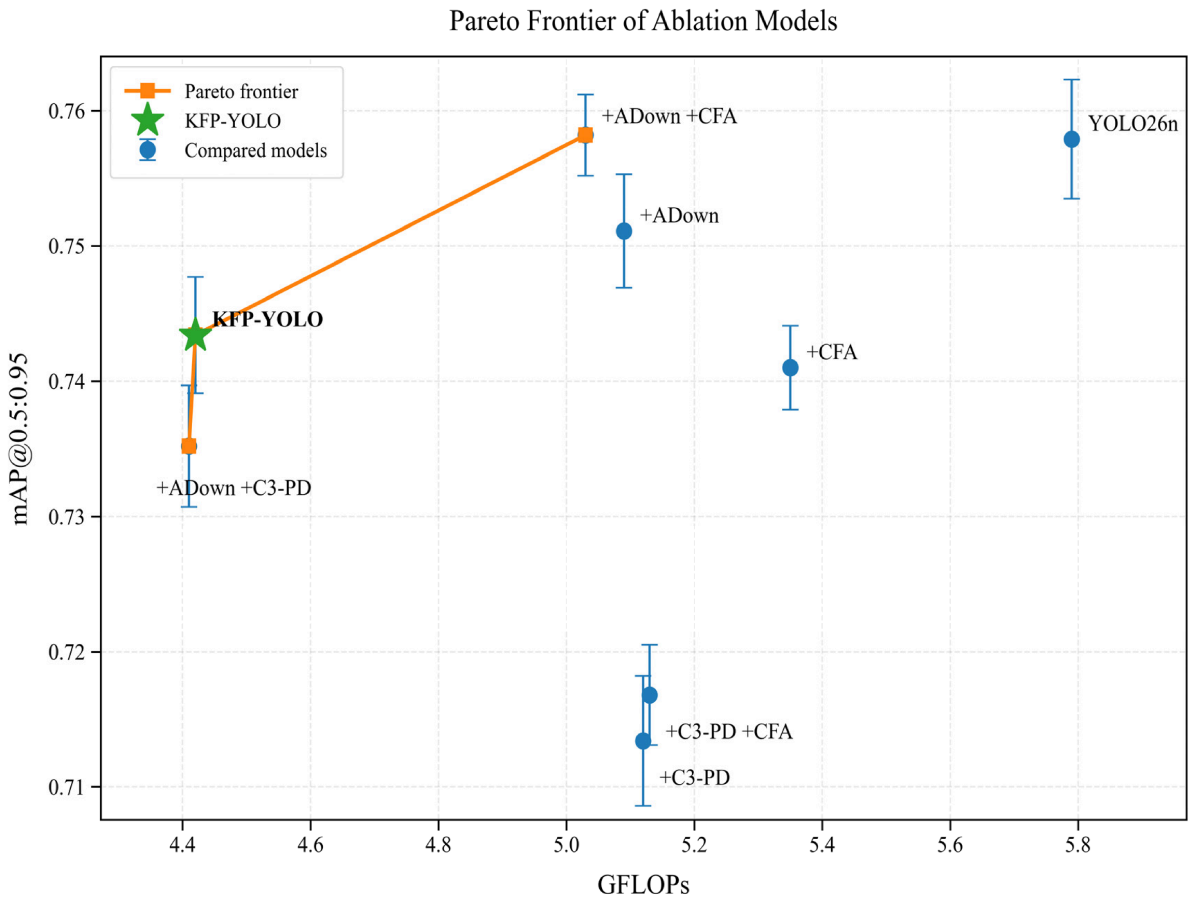
To further evaluate the trade-off between detection accuracy and computational complexity, Pareto frontier analysis was conducted using GFLOPs as the horizontal axis and mAP@0.5:0.95 as the vertical axis, as shown in Figure 8. A model is considered Pareto efficient when no other model achieves equal or higher detection accuracy with equal or lower computational cost. In the mainstream model comparison, the Pareto frontier consists of YOLO26n, YOLOv13n, and YOLOv13s, with corresponding coordinates of (5.79, 0.7579), (6.30, 0.8065), and (22.30, 0.8715), respectively. YOLO26n is located at the low-computation end of the frontier, YOLOv13n achieves a substantial improvement in detection accuracy with a moderate increase in computational cost, and YOLOv13s provides the highest mAP@0.5:0.95 at a considerably higher computational cost. The remaining models are dominated by at least one of these frontier models.

For the ablation experiments, the Pareto frontier consists of ADown+C3-PD, KFP-YOLO, and ADown+CFA, with corresponding coordinates of (4.41, 0.7352), (4.42, 0.7434), and (5.03, 0.7582), respectively. Compared with ADown+C3-PD, KFP-YOLO increases mAP@0.5:0.95 by 0.82 percentage points with an increase of only 0.01 GFLOPs. Although ADown+CFA achieves higher detection accuracy, it requires 0.61 more GFLOPs than KFP-YOLO. Therefore, KFP-YOLO occupies an intermediate position on the ablation Pareto frontier and provides a favorable trade-off between computational complexity and detection performance.



(**a**)

Figure 8. *Cont.*



(**b**)

Figure 8. Pareto frontier analysis of model comparison and ablation experiments: (**a**) Pareto frontier of mainstream model comparison; (**b**) Pareto frontier of ablation experiments.

3.4. External Test Set Validation

To further evaluate model generalization beyond the internal KFP-PDD test subset, all comparison models trained using three random seeds (0, 42, and 134) were evaluated on an independently prepared external test set containing 1052 images and 3377 annotated instances. The external images followed the same 11-class annotation system as KFP-PDD but were not used for model training, validation, hyperparameter adjustment, or model selection. No data augmentation was applied to the external test set, and all models were evaluated using the same input resolution and preprocessing procedure. The corresponding averaged results are presented in Table 7.

Table 7. Performance comparison of different models on the external test set.

Model	Size (MB)	Params (M)	GFLOPs	FPS	mAP@0.5	mAP@0.5:0.95	Precision	Recall
YOLOv5n	5.04	2.51	7.19	321.85	0.8263 ± 0.0033	0.5696 ± 0.0052	0.7847 ± 0.0055	0.7495 ± 0.0035
YOLOv8n	5.98	3.01	8.2	311.78	0.9085 ± 0.0024	0.6863 ± 0.0046	0.8583 ± 0.0084	0.8341 ± 0.0053
YOLO11n	5.23	2.59	6.45	266.95	0.9180 ± 0.0052	0.7078 ± 0.0088	0.9002 ± 0.0104	0.8292 ± 0.0082
YOLO12n	5.28	2.57	6.49	143.02	0.9334 ± 0.0016	0.7486 ± 0.0023	0.8874 ± 0.0023	0.8728 ± 0.0020
YOLOv13n	5.17	2.46	6.3	109.32	0.9351 ± 0.0011	0.7832 ± 0.0015	0.9022 ± 0.0036	0.8951 ± 0.0011
YOLOv13s	18.76	9.55	22.3	73.08	0.9506 ± 0.0005	0.8201 ± 0.0009	0.9377 ± 0.0015	0.9259 ± 0.0006
YOLO26n	5.16	2.51	5.79	259.79	0.9342 ± 0.0024	0.7382 ± 0.0031	0.8935 ± 0.0035	0.8972 ± 0.0025
YOLO26s	19.4	9.96	22.54	167.67	0.9493 ± 0.0004	0.8229 ± 0.0006	0.9121 ± 0.0013	0.9224 ± 0.0009
KFP-YOLO	4.05	1.93	4.42	278.97	0.9325 ± 0.0022	0.7322 ± 0.0030	0.8814 ± 0.0033	0.9077 ± 0.0026

Among the compared models, YOLOv13s achieved the highest mAP@0.5 of 0.9506, whereas YOLO26s obtained the highest mAP@0.5:0.95 of 0.8229. However, these two models required 9.55 M and 9.96 M parameters and 22.30 and 22.54 GFLOPs, respectively. Among the lightweight models, YOLO26n achieved an mAP@0.5 of 0.9342 and an mAP@0.5:0.95 of 0.7382, with a Precision of 0.8935 and a Recall of 0.8972. KFP-YOLO achieved an mAP@0.5 of 0.9325, an mAP@0.5:0.95 of 0.7322, a Precision of 0.8814, and a Recall of 0.9077. Compared with YOLO26n, the mAP@0.5 and mAP@0.5:0.95 of KFP-YOLO decreased by 0.17 and 0.60 percentage points, respectively, whereas Recall increased by 1.05 percentage points. KFP-YOLO also retained its lower model complexity, with 1.93 M parameters and 4.42 GFLOPs, and achieved an inference speed of 278.97 FPS.

Compared with its performance on the internal KFP-PDD test subset, the mAP@0.5 and mAP@0.5:0.95 of KFP-YOLO decreased by 1.04 and 1.12 percentage points, respectively, on the external test set. The relatively limited decline indicates that the model retained competitive detection capability on unseen data, although the lower Precision suggests that false-positive predictions remain an important source of error under external acquisition conditions. Overall, the external evaluation provides additional evidence that KFP-YOLO maintains a favorable balance between generalization performance, model complexity, and edge inference efficiency, while further geographically and temporally diverse field validation is still required.

3.5. Detection Results and Feature Activation Visualization

Figure 9 presents representative detection results and feature activation maps of YOLO26n and KFP-YOLO. Overall, the two models produced similar detection results for several relatively clear targets, including fruit infestation by *Grapholita molesta*, leaf chlorosis, fruit fire blight, and leaf insect damage. The activation maps indicate that both models responded to lesion regions, damaged tissues, fruit boundaries, and surrounding contextual features. In some examples, KFP-YOLO exhibited relatively concentrated responses near the target regions, whereas in other cases, its activation remained distributed across nearby branches, leaves, or background areas. Therefore, the visualization results do not indicate that KFP-YOLO consistently produces more focused feature responses in all scenes.

The representative examples also reveal several limitations. In the third image group, KFP-YOLO missed one leaf spot target in a scene containing multiple overlapping disease regions, whereas YOLO26n detected the corresponding target. In the sixth image group, both models failed to detect the healthy fruit targets. In the seventh image group, the relatively distant leaf targets located in the background were not detected by either model. These failures indicate that target density, occlusion, scale variation, and background depth remain important factors affecting detection reliability. When multiple targets are densely distributed, adjacent objects and overlapping features may reduce the separability of individual instances and increase the probability of missed detections.

The failure cases involving distant background targets further suggest that detection performance may deteriorate as the imaging distance increases and the projected target size decreases. Under longer-distance acquisition conditions, small lesions or pest-damaged regions may occupy only a limited number of pixels, causing their features to be weakened or lost during network downsampling. The present results do not establish a specific distance at which the model becomes ineffective, but they indicate that its applicability may be limited for very small, distant, or densely distributed targets. Higher-resolution inputs, image tiling, small-object-oriented detection layers, and more diverse long-distance training samples should therefore be investigated in future work.

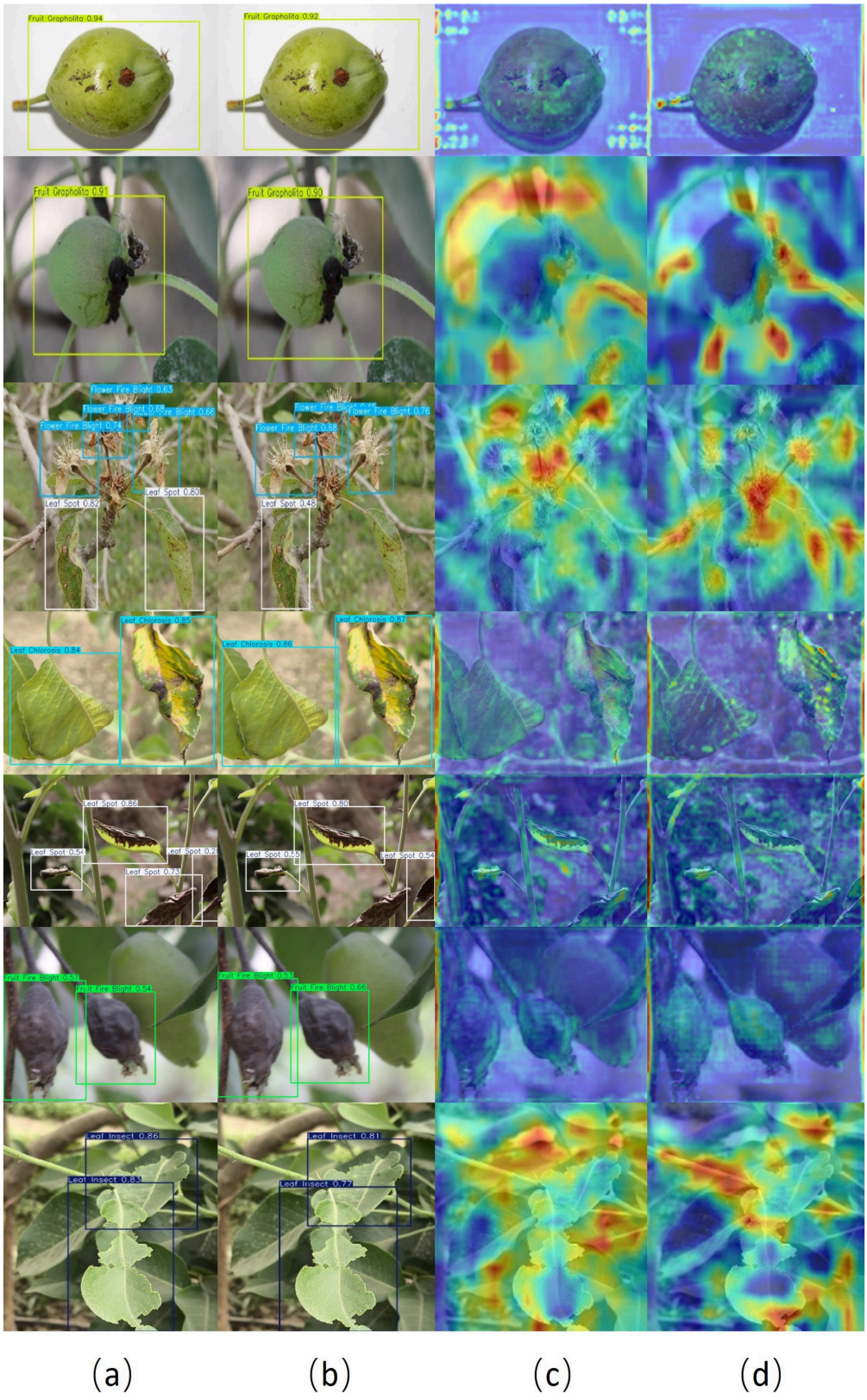


Figure 9. Representative detection results and feature activation maps of YOLO26n and KFP-YOLO: (**a**) detection results of YOLO26n; (**b**) detection results of KFP-YOLO; (**c**) feature activation maps of YOLO26n; and (**d**) feature activation maps of KFP-YOLO.

4. Discussion

This study focused on the joint detection of healthy, diseased, and pest-damaged targets across pear leaves, fruits, and flowers under both controlled and natural orchard

conditions. Compared with datasets dominated by a single organ or acquisition environment, KFP-PDD includes self-collected field images, indoor images, and supplementary public images, thereby increasing the diversity of target scales, backgrounds, illumination conditions, and symptom appearances. However, differences among these acquisition domains may introduce domain shifts. To reduce evaluation bias, the public images were used only for training, whereas the internal validation and test subsets contained self-collected images. An independently prepared external test set was further used to evaluate model generalization under unseen acquisition conditions.

The experimental results show that the main advantage of KFP-YOLO lies in model compression and edge inference efficiency rather than maximum detection accuracy. Compared with YOLO26n, KFP-YOLO reduced the parameter count from 2.51 M to 1.93 M and GFLOPs from 5.79 to 4.42, corresponding to reductions of 23.1% and 23.7%, while increasing the Jetson AGX Orin inference speed from 259.79 to 278.97 FPS. These reductions were accompanied by decreases of 0.29 and 1.45 percentage points in mAP@0.5 and mAP@0.5:0.95, respectively. The ablation results further indicate that the effects of the three modules were not simply additive. ADown mainly reduced the computational cost of the downsampling stages, whereas C3-PD provided stronger model compression and speed improvement but caused a larger decrease in strict localization performance. CFA improved some accuracy indicators, and the ADown+CFA configuration achieved higher detection accuracy than the complete KFP-YOLO model. Nevertheless, KFP-YOLO required fewer parameters and GFLOPs and achieved a higher inference speed than ADown+CFA. Therefore, the final architecture was selected according to the objective of obtaining a more compact model while retaining competitive detection capability, rather than selecting the configuration with the highest detection accuracy alone. The ablation Pareto analysis similarly places KFP-YOLO between the lower-cost ADown+C3-PD configuration and the higher-accuracy ADown+CFA configuration.

Evaluation on the external test set provides additional evidence regarding model generalization. KFP-YOLO achieved an mAP@0.5 of 0.9325 and an mAP@0.5:0.95 of 0.7322, representing decreases of 1.04 and 1.12 percentage points, respectively, relative to its internal test-set performance. Compared with YOLO26n on the external test set, KFP-YOLO showed slightly lower Precision and mAP values but higher Recall, suggesting that the lightweight model retained a relatively high ability to identify true targets while producing more false-positive predictions under external acquisition conditions. The visualization results also show that KFP-YOLO did not consistently generate more concentrated feature responses than YOLO26n. In dense scenes, KFP-YOLO missed one leaf spot target, while both models failed to detect healthy fruits in one example and distant background leaves in another. These failure cases indicate that dense target distributions, occlusion, weak symptom expression, scale reduction, and background complexity remain important sources of error. In particular, when the imaging distance increases and target regions occupy only a small number of pixels, disease and pest features may be weakened during network downsampling, thereby reducing detection reliability.

Under the specified Jetson AGX Orin benchmarking protocol, KFP-YOLO achieved an inference speed of 278.97 FPS using FP16 TensorRT inference, a batch size of 1, MAXN power mode, fixed device frequencies, and 100 warm-up iterations. This result demonstrates the computational efficiency of the lightweight architecture on the selected edge platform. Future work should expand data collection across additional orchards, seasons, devices, growth stages, and environmental conditions, with particular emphasis on long-distance, densely distributed, and severely occluded targets. Higher input resolution, image tiling, small-object-oriented feature layers, and multi-scale training may improve distant-target detection. Further deployment studies should also evaluate latency dis-

tribution, power consumption, energy efficiency, memory usage, temperature variation, and continuous-video detection performance to support the development of orchard-monitoring systems. In addition, the current deployment evaluation was conducted under controlled benchmarking conditions and did not assess several factors associated with long-term commercial operation. Long-term camera maintenance, lens contamination, hardware durability, seasonal changes in canopy structure and illumination, economic feasibility, and integration with commercial orchard-management or precision-spraying systems remain to be validated. These factors may affect system reliability, maintenance requirements, operating costs, and grower acceptance and should therefore be investigated through extended field deployments under commercial production conditions.

5. Conclusions

This study proposes KFP-YOLO, a lightweight object detection model based on YOLO26n, and constructs the multi-organ Korla Fragrant Pear Disease and Pest Dataset (KFP-PDD). By introducing ADown into the backbone, replacing selected C3k2 blocks with C3-PD modules, and incorporating CFA into the multi-scale feature-fusion stages, KFP-YOLO reduces the number of parameters from 2.51 M to 1.93 M and the computational complexity from 5.79 to 4.42 GFLOPs, corresponding to reductions of 23.1% and 23.7%, respectively. Meanwhile, its inference speed on the Jetson AGX Orin platform increases from 259.79 to 278.97 FPS. KFP-YOLO achieves an mAP@0.5 of 0.9429 and an mAP@0.5:0.95 of 0.7434 on the KFP-PDD test set, while obtaining an mAP@0.5 of 0.9325 and an mAP@0.5:0.95 of 0.7322 on the independent external test set. The comparative, ablation, and external-validation results indicate that KFP-YOLO represents a promising lightweight framework that provides a favorable trade-off among detection performance, model complexity, and edge inference efficiency. However, its long-term reliability, transferability, economic feasibility, and system-level applicability should be further validated through long-term, multi-site field deployments across different seasons and production conditions before routine commercial implementation.

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